Unit 1

Functions and Relations

1.1 Sets and Numbers

Introduction to Sets

Definition: Set

A **set** is a collection of objects. If an object is contained in the collection, we say it is an **element** of the set.

Notation:

- We use capital letters (A, B, C, etc.) to represent sets.
- We typically use lower-case letters (x, y, z, etc.) to represent objects.
- To say that an object x is an element of a set A, we write $x \in A$.
- To say that an object x is not an element of a set A, we write $x \notin A$.

Definition: Roster Notation

The simplest way to represent a set is **Roster Notation**, in which we list the elements of the set inside curly braces, separated by commas.

Example. $A = \{1, 2, 3\}$ means that A is the set containing the numbers 1, 2, and 3.

If there are a lot of elements in a set, we can write "..." to indicate a pattern continues.

Example. $B = \{1, 2, ..., 20\}$ means that B is the set containing the _____ numbers _____ through _____ .

Example. $C = \{0, 2, 4, ..., 100\}$ means that C is the set containing the _____ numbers _____ through _____ .

We can also indicate that a pattern continues forever.

Example. $D = \{1, 3, 5, ...\}$ means the set of all _____ .

Example. Let $A = \{1, 2, 3, 4, 5\}$, $B = \{-3, -1, 1, 3\}$, and $C = \{3, 5, 7, 9, 11, ...\}$. Fill in the blanks below with $\in$ or $\notin$ as appropriate.

-1 _____ A	2 _____ A	3 _____ A
-1 _____ B	2 _____ B	3 _____ B
-1 _____ C	2 _____ C	3 _____ C

Standard Number Sets

There are certain sets of numbers that we talk about regularly in algebra.

Name	Symbol		Description
Natural Numbers	$\mathbb{N}$		The counting numbers 1, 2, 3, ...
Integers	$\mathbb{Z}$		The natural numbers, their opposites -1, -2, -3, ..., and 0. Numbers with no fractional part.
Rational Numbers	$\mathbb{Q}$		The numbers that can be represented as fractions $\frac{p}{q}$ where p and q are integers, $q \neq 0$.
Real Numbers	$\mathbb{R}$		The rational numbers, together with irrational numbers to "fill in" the number line.

Note. We will sometimes refer to the **positive integers** to mean the natural numbers, and the **nonnegative integers** to mean the natural numbers plus zero.

Note. The same rational number can be written in many forms, e.g. $\frac{3}{5} = \frac{6}{10} = \frac{-9}{-15}$ etc. These are all the same rational number.

Example. For each number below, determine which of the standard number sets contains it.

	$\mathbb{N}$	$\mathbb{Z}$	$\mathbb{Q}$	$\mathbb{R}$	Notes
6					
$-\frac{16}{4}$					
$\frac{3}{5}$					
-0.78					
$0.\overline{3}$					
$\sqrt{2}$					

How can we recognize rational and irrational numbers?

- All _____ are rational.
- Any numbers written as a _____ is rational.
- Every _____ or _____ decimal is rational.
- All other decimals are irrational.
- Square roots of positive integers are either _____ or irrational.

Terminating and Repeating Decimals are Rational

Theorem

Every fraction can be represented as a terminating or repeating decimal.

We prove this by long division. Here, we will show two examples, but this works in general.

$\frac{5}{8}$ as a terminating decimal.

$$8 \overline{\big)\, 5.000}$$

$\frac{19}{22}$ as a repeating decimal.

$$22 \overline{\big)\, 19.000}$$

Theorem

Every terminating decimal can be written as a fraction.

We see an example, but this works in general. 0.385 can be written as a fraction.

Denominator is Numerator is

$0.385 =$

Theorem

Every repeating decimal can be written as a fraction.

We see an example again. $0.\overline{87}$ can be written as a fraction.

Square Roots are Typically Irrational

Theorem

The square root of a positive integer is either a positive integer or an irrational number.

Proof. We'll argue that $\sqrt{2}$ is irrational. The same argument works for any prime number; a little more detail is needed for numbers that aren't prime.

- Suppose $\sqrt{2}$ were rational. Then it could be written as a fraction $\frac{p}{q}$ in lowest terms. That means that p and q have no _____ .
- $\sqrt{2} = \dfrac{p}{q}$ means $2 = \dfrac{p^2}{q^2}$. But $2 = \dfrac{2}{1} = \dfrac{2q^2}{q^2}$, so $p^2 = 2q^2$.
- That means _____ is a factor of p^2. Since _____ is prime, it must be a prime factor of p. Say $p = 2r$.
- Then $p^2 = (2r)^2 = 2^2 r^2$.
- Going back to the fraction, we have $2 = \dfrac{p^2}{q^2} = \dfrac{\quad}{q^2} = \dfrac{\quad}{q^2} = \dfrac{2q^2}{q^2}$.
- Now we see that $2r^2 = q^2$, so _____ must be a factor of q^2 and thus q.
- So 2 is a _____ of p and q.
- But at the beginning we picked p and q to have no _____ !
- That means we can't pick values p and q like that, so $\sqrt{2}$ can't be written as a fraction—it is _____ . $\square$

Set Builder Notation

We have an alternate way of describing the elements of a set.

Definition: Set Builder Notation

Set Builder Notation is a method of describing a set in terms of a generic element, followed by a description of rules the elements must follow. Set Builder Notation looks like this:

$$\{\text{generic element} \mid \text{rules}\}$$

Example. Let $A = \{2n \mid n \in \mathbb{Z}\}$

- List three elements of A:
- How else could we describe A?

Example. Let $B = \left\{\frac{n}{10} \mid n \in \mathbb{N}\right\}$

- List three elements of B:
- Is $\frac{3}{5} \in B$?
- Are there any integers in B?

Example. Let $C = \left\{\frac{p}{q} \mid p \text{ is even and } q \text{ is odd}\right\}$

- List three elements of D:
- Are there any integers in C?

Properties of Real Numbers

To do math to real numbers, we use two operations: **addition** and **multiplication**. These operations have certain properties.

Commutative Property of Addition For any real numbers a and b, $a + b = b + a$	**Commutative Property of Multiplication** For any real numbers a and b, $ab = ba$
Associative Property of Addition For any real numbers a, b, and c, $(a + b) + c = a + (b + c)$	**Associative Property of Multiplication** For any real numbers a, b, and c, $(ab)c = a(bc)$
Distributive Property For any real numbers a, b, and c, $a(b + c) = ab + ac \qquad (b + c)a = ba + ca$	

You might think of **subtraction** and **division** as operations, but we meet them in these next properties.

Additive Identity Property There is an additive identity 0 such that, for all real numbers a, $a + 0 = 0 + a = a$	**Multiplicative Identity Property** There is a multiplicative identity 1 such that, for all real numbers a, $1a = a \cdot 1 = a$
Additive Inverse Property Every real number a has an additive inverse $-a$ such that $a + (-a) = -a + a = 0$	**Multiplicative Inverse Property** Every real number $a \neq 0$ has a multiplicative inverse $\frac{1}{a}$ such that $a \cdot \frac{1}{a} = \frac{1}{a} \cdot a = 1$

We define subtraction and division using the inverses:

- $a - b =$ _____
- $a \div b =$ _____

Simplifying Expressions: Basic Moves

Recall:

Definition

- An **expression** is a sequence of numbers, variables, and symbols that can be resolved to represent a number.
- To **evaluate** an expression is to determine what number it represents.
- Two expressions are said to be **equivalent** if they represent the same number, no matter what any variables represent. To **rewrite** an expression is to write an equivalent expression in a different form.
- To **simplify** an expression is to write an equivalent expression in some standard "simplest" from.
- The **terms** of an expression are parts of an expression combined using *addition*. The **factors** of an expression are parts of an expression combined using *multiplication*.

There are a number of common steps we use to simplify expressions.

- Cancel common factors in the numerator and denominator of a fraction.
- Use the commutative property to rearrange terms and factors.
- Multiply numbers within terms.
- Use the distributive property to eliminate parentheses around addition (or _____).
- "Combine Like Terms" (using the distributive property).
- Use properties of exponents to combine factors.

Example. Simplify $3(x+4) - 5(x-3)$.

Step	Why?

Properties of Equality

Definition: Equation

An **equation** is a statement that two expressions are equal. It is of the form

$$\text{expression} = \text{expression}$$

The equation is *true* if the two expressions represent the same number.

- A **solution** to an equation is a value(s) of the variable(s) in the equation that makes the equation true.
- The **solution set** of an equation is the set of all solutions of the equation.
- Two equations are **equivalent** if they have the same solution set.
- To **solve** an equation means to find its solution set.

We use equations in several ways.

- To show equivalent expressions.

  Example: _____

- To define a variable symbol.

  Example: _____

- To describe an unknown quantity.

  Example: _____

The following properties of equality allow us to get equivalent equations.

Additive Property of Equality	**Multiplicative Property of Equality**
For any expressions X and Y and any real number c, the equations $$X = Y \quad \text{and} \quad X + c = Y + c$$ are equivalent.	For any expressions X and Y and any real number $c \neq 0$, the equations $$X = Y \quad \text{and} \quad c(X) = c(Y)$$ are equivalent.

Symmetric Property of Equality

For any expressions X and Y, the equations

$$X = Y \quad \text{and} \quad Y = X$$

are equivalent.

Solving Equations: Basic Moves

The properties of equality give us two basic moves that we can use to solve equations.

- We can _____ the same number to each side of an equation.

  We can also _____ (this is just adding the opposite).

- We can _____ by the same number on each side of an equation.

  We can also _____ (this is just multiplying by the inverse).

Note. We can _____ any number, so it is safe to _____ variable expressions. However, we must be careful not to _____ by zero, so we don't _____ by variable expressions at this stage.

We choose basic moves in order to isolate the variable on one side.

Example. Solve $2x - 3 = 7$

Step	Why?

Example. Solve $5x - 7 = 2x + 5$

Step	Why?

Example. Solve $\frac{2}{3}x - \frac{3}{5} = \frac{1}{6}x + \frac{4}{5}$

Step	Why?

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